The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.5.4 (prototype)
Log entry date	12 November 2026
Status	Released: deployed by the merge of pull request #11 into the GitHub main branch, together with the 0.5.4 documents and this log. Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch
Previous version	0.5.3, 12 November 2026, pull request #10 (commit 2bda25f), which also added the Veldmeer game
Prepared by	A. Paarlberg

1. Purpose of This Log

This Development Log records the changes from version 0.5.3 to version 0.5.4 of The Spaghetti Engine. Version 0.5.4 makes the tool easier to use on a normal laptop or desktop screen. It enlarges the text, simplifies the layout, makes the Network tab quicker for building a model, and makes the figure editor produce figures that can be placed directly in a scientific paper. It also documents the Veldmeer game, which was released with version 0.5.3 but was not yet described in the documents.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.5.4

Area	Change
Text and spacing	Larger base text (16 px for body text) and fewer, calmer elements on every page. At 1366 by 768 pixels the controls above the work area take under 60 px instead of about 290 px.
Top bar	A model menu (My models, Import JSON, Export JSON, Export Analysis Workbook), the save status, an Advanced tools switch, and a Help menu.
Side menu	The tab rows are replaced on laptop and desktop screens by a side menu in which each step folds open to show its tabs. The analysis scope is shown as one line with a Change scope button.
Help menu	Getting started guide, the Veldmeer game, About & documentation, and a short guide to building a map with the keyboard shortcuts, which previously filled the empty Inspector.
Network tab	A full-height map, a one-row toolbar, an Inspector that appears only for a selection, New effect and New cause buttons, placement of new concepts in view, and a hint line with the next step.
Figure editor	A journal style by default, cleaning for print, four numbered steps, readiness checks under the preview, and a Screen style for slides.
Documentation	Version 0.5.4 of the User Manual and the Methodological Annex, each with an update chapter, and this log. The About page describes the Help menu and the game.

3. The Changes in Detail

3.1 Layout and text

Text. The text scale was raised by one step throughout: small text from 12 to 14 px, body text from 14 to 16 px, and fixed sizes in figures and panels by 1 to 2 px. Tooltips use 14 px.

Top bar. The version number moved into the tooltip of the name. The model actions moved into one menu. Basic and Advanced became one switch, Advanced tools.

Side menu. Each step of the work is a button that folds its tabs open or closed; the current step is open. The scope bar is hidden unless part of the model is selected or the user asks for it. The Getting started guide no longer occupies the top of each tab, and the footer is reduced to one line. Below laptop width the previous tab rows are kept.

3.2 Building a model on the Network tab

Canvas and toolbar. The map takes the height of the screen. Present and Export PNG are icons below 1536 px width, and the choice of full model, module overview or open module moved into the View menu, so that the toolbar fits on one row.

Inspector. The Inspector appears only when a concept or relationship is selected and has a close button. For a concept, it shows the name, the module, a Connect box (New effect, New cause, and a folded link to an existing concept), the relationships, and a folded section with the definition and starting value. For a relationship, it shows the effect, the strength as four buttons, and the exact weight and description folded.

Placement. Add concept places the card in the centre of the view at a free spot. New effect and New cause place the new card beside the selected one, connect it, and pan the map if the card would be off screen.

3.3 Figures for papers

Styles. Journal (Arial 7 pt, Okabe-Ito, PNG at 600 dpi), Serif (Times New Roman 8 pt, Tol bright, PDF) and Screen (as shown, with title, 300 dpi). The Journal style is the default; the stored preferences of earlier versions are not reused, so every user starts from it.

Clean for print. Removes screen notes, decorative panel backgrounds and light gridlines, sets grey text to black, and widens lines to at least 0.25 pt. The rules are given in section U.1 of the Methodological Annex.

Panel and checks. The panel has four numbered steps (Style, Width in the paper, Panel letter and title, File). All other settings are under Fine-tune. Readiness checks under the preview report text size, line width, resolution, column width, palette and title.

4. Implementation Choices

- **One source of navigation.** The side menu and the mobile tab rows are generated from the same list of steps and tabs, so the two cannot diverge.
- **Notes and grids are marked, not guessed.** Figure notes and chart grids are marked when the figure is captured, because the class names that identify them are removed when styles are inlined. Unmarked elements are recognised only by conservative geometric and colour rules.
- **No change to the engine.** All changes are in the interface and in the export pipeline. The model format is unchanged, so models saved with earlier versions open without conversion.

5. Effect on Existing Results

None. No computation was changed, and every analysis gives the same numbers as in version 0.5.3. Exported figures look different by default (journal style instead of the screen style), but the plotted values are identical. The Screen style reproduces the earlier appearance.

6. Defects Found and Fixed

Defect	Fix
The Network toolbar wrapped onto two rows on a laptop screen.	Icon buttons below 1536 px and the module choice moved into the View menu.
A new concept could appear on top of another card or outside the view.	New concepts are placed at a free spot in view, and the map pans to them when needed.
Screen notes stayed in exported figures, because their class was lost before cleaning.	Notes and grids are marked with data attributes before the classes are removed.
The help text for building a map was visible only while nothing was selected.	It moved into the Help menu, where it is always available.

7. Verification

- The six automated tests pass.
- Every tab was opened in a browser at 1366 by 768 pixels and at phone width (390 px) with no errors in the console and no sideways scrolling. Before and after screenshots were kept with the project files.
- A model of four concepts and three relationships was built with Add concept, New effect and New cause only, with no errors.
- The baseline figure was exported as PNG (4110 by 2964 pixels, 600 dpi recorded in the file), PDF and SVG. The journal output has no panel backgrounds, notes or gridlines, and all text is black.

8. Deployment Status

Released through pull request #11, merged into the main branch on 12 November 2026. Netlify deployed the site from the main branch. The deploy preview of the pull request was checked before the merge.

9. Prototype Scope and Known Limitations

- Long concept names in the baseline figure are shortened with an ellipsis to fit the layout of the figure.
- The readiness checks follow common journal requirements; specific journals may differ.
- The side menu is designed for screens of at least 1024 px width; smaller screens use the tab rows.

10. Related Documentation

The Spaghetti Engine, User Manual (version 0.5.4), chapter "Update for Version 0.5.4". The Spaghetti Engine, Methodological Annex (version 0.5.4), chapter "Update for Version 0.5.4". Development Log version 0.5.3.

11. Next Steps

- Wrap long concept names in figures instead of shortening them.
- Integrate the update chapters into the body of the User Manual and the Methodological Annex at the next full revision.